

Tejaswi Bagepally

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PROFILE

Master's student in Applied AI with nearly two years of production AWS data engineering for a US banking client (Glue, Redshift, CloudWatch, CI/CD) and hands-on ML deployment (LSTM, XGBoost, TensorFlow Lite on Lambda).

EDUCATION

Deggendorf Institute of Technology *10/2025 – Present*
M.Eng. Applied AI for Digital Production Management Germany
Focus: Machine Learning and Deep Learning, Advanced Statistical Methods and Optimisation, Data Management, Advanced Intelligent Systems.

MLR Institute of Technology *08/2019 – 08/2023*
B.Tech Computer Science Engineering (German grade: 2.02) Hyderabad, India

PROFESSIONAL EXPERIENCE

Virtusa, Hyderabad — Associate Engineer (Data Engineering, AWS) *01/2024 – 09/2025*
Project: Data warehouse migration to AWS for City National Bank (USA); team of 3–5

- Traced the data lineage of the bank's legacy reporting views back to their source tables.
- Migrated reporting tables and views from Amazon RDS to Redshift, rebuilding SQL logic and validating parity.
- Built and enhanced AWS Glue ETL jobs loading data into Redshift according to reporting business logic.
- Refactored slow Redshift views with CTEs and restructured joins to reduce query runtimes for reporting.
- Monitored Glue jobs with Amazon CloudWatch and maintained CI/CD pipelines that automated deployments.
- Built an internal serverless proof of concept (S3, Lambda, DynamoDB) handling tens of thousands of records/day.

Technologies: Amazon RDS, Redshift, AWS Glue, CloudWatch, S3, Lambda, DynamoDB, SQL, CI/CD

Virtusa — Intern *01/2023 – 04/2023*
Hands-on work with AWS compute, storage, networking and security services; supported cloud learning labs.

PROJECTS

Remaining Useful Life Prediction for Turbofan Engines
Python, TensorFlow, XGBoost, TensorFlow Lite, AWS Lambda, API Gateway, Streamlit

- Trained LSTM and XGBoost models on all four subsets of the NASA CMAPSS dataset (709 training engines, about 160,000 engine cycles, 21 sensors) to predict Remaining Useful Life.
- Reduced RMSE from 19.19 cycles (XGBoost baseline) to 13.17 cycles with the LSTM and improved the NASA Score from 64,337 to 9,230.
- Engineered 45 rolling-window features; deployed the LSTM as TensorFlow Lite on AWS Lambda.
- Built a live Streamlit dashboard with a real-time RUL gauge and fleet monitoring for 20 engines.

Production Route Optimisation with Data Mining (Medium article)
R, arules (Apriori), igraph, A search*

- Mined association rules with Apriori (support ≥ 0.30, confidence ≥ 0.70) from 30 board transactions across 9 PCB production stations, recovering the process flow and its 33.3% rework rate.
- Modelled the line as a weighted directed network and ran A* route search; showed a Braess-type paradox where a shortcut skipping testing lowers route cost but would ship 1 in 3 boards defective.

PUBLICATION

Paper on a multilingual OCR converter (English, Telugu, Hindi), IEEE ICCCI 2023. IEEE Xplore

TECHNICAL SKILLS

- **Programming:** Python, SQL, R, Java
- **Machine Learning:** TensorFlow, XGBoost, LSTM networks, time-series modelling, feature engineering, model evaluation, exploratory data analysis
- **Generative AI (concepts):** LLMs, retrieval-augmented generation, embeddings, fine-tuning
- **Cloud & Data:** AWS (Redshift, Glue, RDS, S3, Lambda, DynamoDB, CloudWatch), ETL, data migration
- **Tools & Deployment:** CI/CD, TensorFlow Lite, Streamlit

CERTIFICATIONS

AWS Certified: Solutions Architect – Associate · Data Engineer – Associate · AI Practitioner

LANGUAGES

English (Fluent) — German (A2, actively improving) — Hindi (Fluent) — Telugu (Native)

Tejaswi Bagepally Lappersdorf, October 25, 2026